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Cybersecurity Matters for Primary School Students: A Scoping Review of the Trends, Challenges, and Opportunities

Yukun Xu and Hui Li

***Abstract*—Primary school students, despite their vulnerability to cyberattacks, lack targeted cybersecurity education. Using Scopus and Google Scholar, this scoping review analyzed 15 articles (2014-2024) following PRISMA-ScR guidelines to examine the landscape of cybersecurity education for this age group. Four key themes emerged: educational tools and methods, theoretical perspectives on online risks, parental engagement, and children's online behaviors and risks. Findings revealed that while students possess some awareness, their understanding is often superficial, leading to overconfidence and risky online practices. A disconnect was observed between parents and teachers regarding responsibility and effective safety practices. Furthermore, limited implementation and outdated resources hinder the effectiveness of promising pedagogical approaches like digital games and interactive platforms. This review highlights the urgent need for comprehensive cybersecurity education fostering critical thinking and stakeholder collaboration. It provides practical implications for educators, parents, and policymakers to promote a culture of online safety alongside recommendations for future research.**

***Index Terms*—Cybersecurity Education; Digital Literacy; Online Risks; Parental Engagement; Primary School Students**

I. INTRODUCTION

THE pervasive use of digital technology has significantly transformed society, particularly impacting children. According to the National Center for Education Statistics [1], the percentage of 3- to 18-year-olds with internet access at home—via computers or smartphones—rose substantially from 2016 to 2021, reaching 97%. This widespread connectivity, while beneficial, also exposed children to numerous potential online risks. Primary school students, in particular, are vulnerable to cyberbullying and various cyberattacks, as they are often novice users of digital technology and still developing their understanding of online safety. Although recent studies have begun to examine cybersecurity issues among K-12 students, there has been a lack of comprehensive literature reviews that address the specific challenges encountered by primary students, along with potential solutions for early intervention. This scoping review sought to bridge this research gap by critically analyzing studies published over the past decade that focused on cybersecurity concerns for primary school students. Through this review, the article aimed to elucidate key areas of focus and identify strategies for enhancing cybersecurity education in this vulnerable demographic.

A. Cybersecurity: Protecting Schools and Educating Students

Cybersecurity is defined by the Cybersecurity and Infrastructure Security Agency (CISA) of the United States [2] as the "art of protecting networks, devices, and data from unauthorized access or criminal use" while ensuring the confidentiality, integrity, and availability of information. Cybersecurity is vital for ensuring individuals' safety in the digital realm. As society increasingly relies on digital tools, the importance of cybersecurity continues to rise. This trend is reflected in Google Trends, indicating that searches for "cybersecurity" peaked between November 2019 and November 2024. However, searches for "cybersecurity" constituted approximately 1% of the searches for "AI" during the same timeframe, indicating a significant disparity in public engagement between these two technological domains.

In the educational context, cybersecurity initiatives can be classified into two main categories: protective measures for schools and educational efforts to raise student awareness. On the one hand, schools are taking proactive steps to secure themselves against potential online risks. For example, CISA [3] offered extensive online toolkits designed for schools and school districts that outlined procedures to follow in the event of cyber incidents, ensuring educators can respond effectively. On the other hand, there is a growing emphasis on promoting cybersecurity education among students. This approach, referred to as "cybersecurity education," involves various methods employed by governments and educational institutions to inform children about the potential online risks they may encounter. For example, CISA [4] provided engaging online lessons and games specifically designed for students in grades K-5, along with resources for educators and parents to enhance their understanding of cybersecurity. Additionally, the Los Angeles Unified School District [5] offered a range of online activities for K-12 students aimed at teaching them how to navigate the digital landscape safely. Together, these efforts highlighted the dual approach of reinforcing cybersecurity

Yukun Xu is a current master's student in the Learning Design and Technology program at Stanford University (e-mail: xuy30@stanford.edu).

Hui Li is the corresponding author of this article. He is the Chair Professor at the Department of Early Childhood Education at the Education University of Hong Kong (e-mail: huili@eduhk.hk). He is also Honorary Professor at Macquarie University and Shanghai Normal University.

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measures within schools and fostering a culture of cyber awareness among young learners.

B. The Existing Review Studies of Cybersecurity Education

Cybersecurity is a multifaceted field that encompassed various domains, including social engineering, construction, and machine learning [6], [7], [8]. Researchers made efforts to categorize existing studies on cybersecurity, identifying notable areas of interest. For example, Suryotrisongko and Musashi [9] reviewed literature published from 2015 to 2019 and classified cybersecurity research into eight categories: (1) applied cybersecurity, (2) cybersecurity data science, (3) cybersecurity education and training, (4) cybersecurity incidents, (5) cybersecurity management and policy, (6) cybersecurity technology, (7) human and social cybersecurity, and (8) theories in cybersecurity.

Within the educational realm, there was an increasing recognition of the necessity to impart cybersecurity knowledge across all age groups. Blažič [10] emphasized the need for updates in cybersecurity education among European employees, while Sinha [11] explored various theories and approaches that educational institutions adopted towards cybersecurity. Ulven and Wangen [12], focusing specifically on higher education, identified nine risks faced by institutions, underlining the complex nature of cybersecurity threats. Li and Liu [13] further underscored the urgency of cybersecurity education by highlighting the rise in cyberattacks, particularly those leveraging advanced technologies such as AI and machine learning. Their study also noted the lack of consensus on defining cyberattacks, complicating protection efforts by governments and judicial systems.

The demand for enhanced cybersecurity education is evident in multiple studies that targeted various aspects of this domain. For instance, Buchan et al. [14] investigated the privacy awareness and digital fluency of Canadian youth, finding disparities influenced by geographic location, socioeconomic status, and access to technology. In contrast, Brown et al. [15] reviewed ethics education in technology, noting a lack of consensus on definitions of ethics within the literature. They identified the challenge of integrating new topics, such as ethics education, into already packed curricula. Similarly, Ibrahim et al. [16] conducted a literature review emphasizing the scarcity of research focusing on cybersecurity education for primary students, noting that previous studies predominantly centered on secondary education. They highlighted the absence of a standardized curriculum on cybersecurity education worldwide, complicating instructional efforts.

While there was limited research specifically targeting primary education, this age group remained crucial in the discourse on cybersecurity. Eyal and Te'eni-Harari [17] observed that younger participants, particularly elementary-aged children, demonstrated more consistent positive outcomes in digital media literacy interventions compared to middle and high school students, who exhibited mixed results. This pattern suggested a pressing need to prioritize cybersecurity education within primary education. Consequently, this scoping review aimed to address these gaps by analyzing existing literature on cybersecurity education for primary students. The following research questions guided this investigation:

1) What key themes and categories emerge from the literature on cybersecurity education for primary school students?
2) What specific online risks and threats do primary school children encounter in their digital interactions, and how do these risks vary by demographic factors?
3) What effective pedagogical approaches and resources are being utilized to enhance cybersecurity awareness and skills among primary school students?

II. METHOD

This study aimed to review the current literatures related to cybersecurity in children aged 6- to 12-year-old. The guidelines of the PRISMA extension for scoping reviews (PRISMA-ScR) statement was followed for this review.

A. Search Strategy

This scoping review used Scopus and Google Scholar as the database and the following search terms in combination: ("digital security" OR "cybersecurity" OR "information security") AND ("elementary" OR "primary" OR "youth" OR "children"). To explore the scope of cybersecurity, this review included peer-reviewed articles published in the last 10 years (Jan. 1, 2014-Dec. 31, 2024).

B. Inclusion and Exclusion Criteria

This review followed four inclusion criteria: (1) the articles included "elementary", "primary", "youth", or "children" in their titles; (2) the articles focused on children that are elementary school-aged, which typically include children aged 6 to 12, and related-stakeholders; (3) the articles focused on cybersecurity or related awareness education; (4) the articles are published in English. The specific criteria are presented in Table 1. The first two criteria were included to better discern the age range of participants focused in the study. The third criterion pinpointed the focus of this study, and the fourth criterion ensured the readability of the articles to the reviewer.

To better explore the scope of existing articles, articles were removed using the following criteria: (1) the articles focused on children with special needs; (2) the articles were secondary data analyses; (3) the articles didn't specify the age range of the target children; (4) the articles focused on the process of teacher education. After reviewing a few articles, the first criterion was added: with participants focused on children with special needs, studies generally divert their attention towards testing the effectiveness of a technological assistance tool, which differs from the focus on cybersecurity. To avoid misunderstanding the data, the second criterion was added. The final two criteria allow the focus of this review to be on primary school children and the adults around them.

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TABLE I
INCLUSION/EXCLUSION CRITERIA

Criterion	Inclusion	Exclusion
Scope of research	Empirical studies	• Not empirical studies • Secondary data analyses • Literature reviews, editorials, position statements, perspectives
Type of documents	Peer-reviewed scholarly journal articles	• Not Peer-reviewed scholarly journal articles
Language	English	• Other languages other than English
Educational level	Children aged from 6-12	• No age mentioned in the article • Children aged over 12 years old • Children with special need
Period	January, 2014 - December, 2024	• Prior to 2014
Research topic	Digital security and related awareness education	• Articles that did not focus on the children's or parents' digital security skills

C. Study Selection

A total of 3,475 articles were found from Scopus and Google Scholar. First, 2 duplicate articles were removed using Zotero. Then, 3,404 articles were excluded because they (1) focused on a different age range than the targeted age (n = 497); (2) focused mainly on teacher education (n = 41); (3) were literature reviews (n = 98); and (4) didn't include "elementary", "primary", "youth", or "children" in their titles (n = 2,768). After examining the abstract and method of the remaining 69 articles, 54 articles were further removed because they (1) didn't primarily focus on elementary-aged children or didn't specify the targeted age range (n = 25); (2) focused mainly on teacher education (n = 1); (3) didn't focus on cybersecurity (n = 26); (4) included repeated information on app development as that of a selected article (n = 1); and (5) conducted a secondary data analysis (n = 1). Since multiple articles did not solely focus on elementary-aged children, the reviewer looked for descriptions of the participants' ages: if more than half of the children addressed in the study were below 12-year-old, the study was selected. If a study discerned between primary school children and older children, the study would also be selected. In the end, 15 articles were selected and analyzed in this study. The flow diagram is presented in Fig. 1.

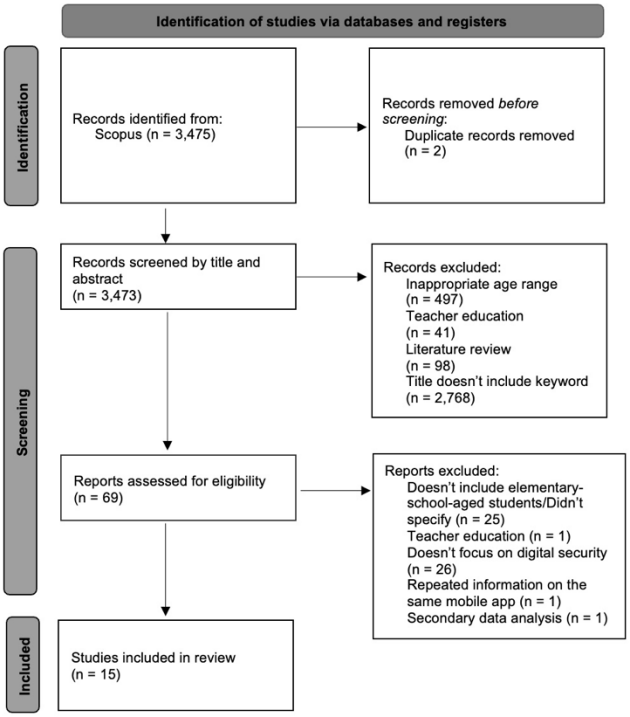


Fig. 1. PRISMA flow diagram of literature search and data charting.

III. RESULTS

Overall, studies on cybersecurity from the perspective of elementary-age students were rare, as research predominantly focused on middle and high school students and university populations. Among the few studies centered on elementary students, seven were conducted in European countries (i.e., the UK, Greece, Spain, Netherlands, Poland, and Norway), two studies were conducted in the United States, three studies took place in Malaysia, and Cyprus contributed two studies. Israel and Saudi Arabia each produced one study on cybersecurity. Full information of the studies is provided in the Appendix.

Although the majority of the selected studies used quantitative or mixed methods in data collection and analysis, as presented in Fig. 2 and Fig. 3, studies on cybersecurity issues predominantly sampled in small sizes. Fig. 3 shows a positively skewed graph and a higher concentration of studies sampling less than 100 participants.

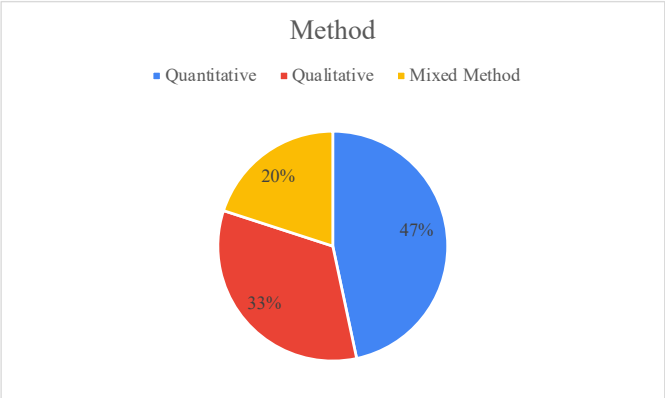


Fig. 2. Distribution of methods.

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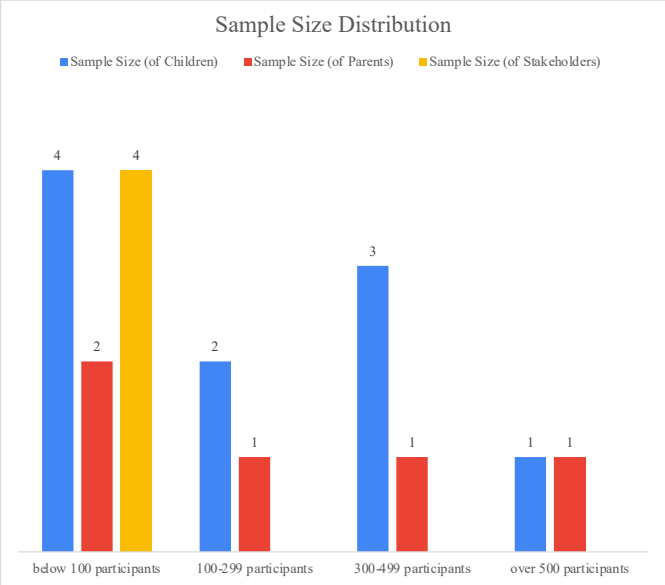


Fig. 3. Distribution of sample size.

A. Themes of Cybersecurity

Because no established theories or coding schemes existed for this specific topic, we employed a three-step, iterative process using ChatGPT-4o to assist with thematic analysis. First, in May 2024, we conducted an initial analysis using ChatGPT-4o. The titles, abstracts, and keywords of the selected articles were input into the model, which was prompted to categorize them into three to six thematic groups. ChatGPT-4o initially identified four categories: "E-Safety Awareness and Digital Literacy," "Parental Involvement and Concerns," "Cyberbullying and Online Risks," and "Educational Tools and Approaches." While the latter two categories were clearly defined, the distinction between "E-Safety Awareness and Digital Literacy" and "Cyberbullying and Online Risks" required further refinement.

Second, to clarify this ambiguity, we engaged in a dialogue with ChatGPT-4, exploring the nuances of each proposed theme. This iterative feedback process led to a tentative distinction based on attitude, with "Cyberbullying and Online Risks" encompassing articles addressing negative digital experiences and "E-Safety Awareness and Digital Literacy" representing a more proactive approach to cybersecurity.

Third, upon further analysis by the co-author, an expert in the field, it became evident that this attitude-based distinction was not consistently reflected in the articles. Consequently, we reorganized the articles, grouping two that focused on validating existing theories and placing the remaining four in a separate category. We then consulted ChatGPT-4 to generate suitable names for the four finalized themes: "Educational Tools and Methods," "Student Behaviors and Online Risks," "Parental Engagement and Concerns," and "Theoretical Perspectives on Online Risks." Figure 4 presents the specific prompts used during this consultation. Tables 2 and 3, respectively, provide the definitions and distribution of articles across these themes. Throughout this process, while ChatGPT-4o aided initial exploration, both authors collaborated to rigorously validate and refine the final coding framework and analysis.

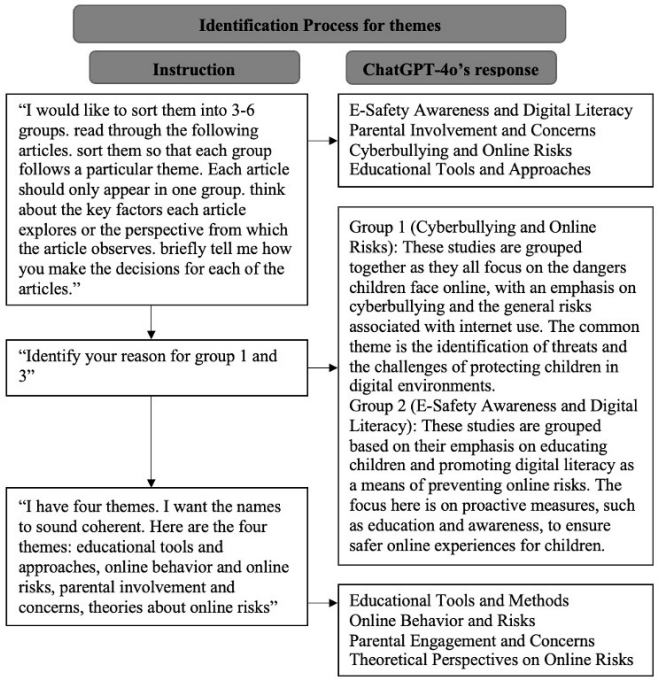


Fig. 4. Conversation with ChatGPT-4o regarding theme identification.

TABLE II
DEFINITION OF THEMES

Educational Tools and Methods	Theoretical Perspectives on Online Risks	Parental Engagement and Concerns	Student Behaviors and Online Risks
The study evaluated tools and methods that are used for teaching primary school children about cybersecurity issues.	The study assessed or added to existing theories on how primary school children perceive and respond to online risks.	The study assessed how parents are involved with educating primary school children about and protecting them from online risks.	The study assessed the types of online risks that primary school children faced.

TABLE III
THEME DISTRIBUTION OF REVIEWED ARTICLES

Educational Tools and Methods	Theoretical Perspectives on Online Risks	Parental Engagement and Concerns	Student Behaviors and Online Risks
Giannakas et al., 2019	Teimouri et al., 2018	Alashwali & Alashwali, 2022	Annansingh & Veli, 2016
Nicolaidou & Venizelou, 2020	Aizenkot, 2022	Larrañaga et al., 2022	Witsenboer et al., 2022
Prior & Renaud, 2020		Abdul Aziz et al., 2023	Martin et al., 2023
Panskyi & Korzeniewska, 2023		Quayyum, 2025	Yusuf et al., 2023
Blinder et al., 2024			

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B. Educational Tools and Methods

Five research articles [18], [19], [20], [21], [22] were classified under "Educational Tools and Methods". Each of these articles identified elements that exhibited educational value or potential in the realm of cybersecurity. Specifically, Giannakas et al. [19] focused on a digital game, CyberAware, which was designed to educate users about privacy and security effectively. Nicolaidou and Venizelou [20] evaluated children's improvement in privacy, security, and protection from cyberbullying through a specific learning platform, finding significant enhancements in the treatment group compared to a control group. Prior and Renaud [22] assessed the educational role of storybooks and online resources but concluded that neither adequately provided current information on password security, thus limiting their educational efficacy. Focusing on the impact of the COVID-19 pandemic, Panskyi and Korzeniewska [21] examined children's abilities to respond to e-threats, discovering that increased awareness during the pandemic had led to improved recognition and resolution of cybersecurity issues. Lastly, Blinder et al. [18] employed a "Would You Rather" game to engage participants in discussions about privacy and security, positing that interactive activities promote understanding of cybersecurity while also supporting students' social-emotional learning (SEL). Collectively, these five articles explore diverse educational strategies and tools that educators may consider for future implementation.

C. Theoretical Perspectives on Online Risks

Only two of the 15 reviewed articles explicitly grounded their research in established theoretical frameworks, highlighting a significant gap in the literature. Teimouri et al. [24] utilized Protection Motivation Theory to investigate how upper-elementary and middle school students respond to online threats. Their study examined the interplay of internal factors, such as self-efficacy and perceived susceptibility to threats, and their correlation with online behaviors. The authors concluded that integrating Protection Motivation Theory, alongside the health belief model, could be instrumental in mitigating online risks for children. While their sample included students older than primary age, the findings suggest potential avenues for understanding how motivational factors influence online safety behaviors in younger children as well.

Aizenkot [23] employed Routine Activity Theory to analyze external factors contributing to cyberbullying victimization among elementary, middle, and high school students. By identifying significant correlations between variables like exposure to motivated offenders, online guardianship, and target suitability, the study provided valuable insights for supporting at-risk youth. Specifically, the findings underscore the importance of online guardianship and reducing exposure to potentially harmful individuals.

While both studies offer crucial theoretical lenses for understanding online risks, their inclusion of older student populations limits their direct applicability to primary school students. Further research is needed to explore how these theories manifest specifically within the context of younger children's online experiences. For instance, investigating how developmental factors influence the relationship between perceived risk severity and protective behaviors (Protection Motivation Theory) or the efficacy of different guardianship strategies (Routine Activity Theory) in the primary school context could significantly enhance our understanding and inform the design of more effective interventions. This theoretical grounding is essential for developing age-appropriate cybersecurity education that addresses the unique needs and vulnerabilities of this age group.

D. Parental Engagement and Concerns

Four articles [25], [26], [27], [28] were categorized under the theme "Parental Engagement and Concerns". Alashwali and Alashwali [26] examined Saudi parents' responses to their children's app usage, finding that these parents actively protect their children's privacy and security while educating them about digital safety. Larrañaga et al. [27] explored Spanish parents' perceptions of cyberbullying, identifying a positive attitude toward their abilities to assist their children but revealing an underestimation of the risks involved. Abdul Aziz et al. [25] studied Malaysian parents' monitoring practices concerning their children's digital engagement, concluding that while parents established monitoring rules, they sought additional support from schools and governmental resources. Quayyum [28] presented an alternate view by using story prompts to illustrate that children expect parental guidance in navigating risks and implementing preventative strategies. Together, these four articles highlight various perspectives on the role of parents in fostering cybersecurity.

E. Student Behaviors and Online Risks

A total of four articles [29], [30], [31], [32] were classified under the theme "Student Behaviors and Online Risks." In their study, Annansingh and Veli [29] analyzed children's digital behavior and identified online risks perceived by students, drawing attention to instances where children lacked adult supervision while online. Martin et al. [30] interviewed teachers in the United States to identify risky behaviors exhibited by students and inappropriate content encountered when unsupervised, as well as instructional strategies that could be implemented to mitigate these issues. Witsenboer et al. [31] used surveys and interviews to assess cybersecurity awareness among different age groups, concluding that elementary students demonstrated caution in some areas but required more guidance regarding specific risks. Most recently, Yusuf et al. [32] focused on the inappropriate content and risks faced by Malaysian children online, noting that a majority of the students surveyed were supervised by parents during their digital interactions. Collectively, these articles examined children's online engagement and the associated risks they encountered.

F. Analysis of Themes

Overall, similar attention was spread on all four themes

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across the nations, as shown by Fig. 5. However, a slight emphasis on theoretical perspectives was shown among Asian and Middle Eastern nations, whereas European countries and the US leaned more toward describing existing risks and experimenting with different educational approaches.

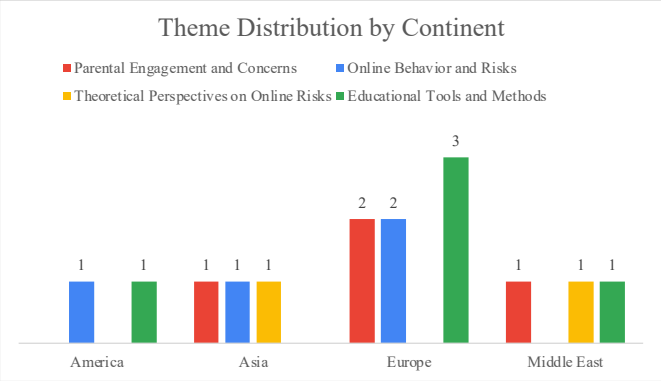


Fig. 5. Theme distribution by continent.

The identified themes also showed correlation to the age range of identified participants (Fig. 6). For example, studies concerned with parental engagement and theoretical perspectives on cybersecurity issues involved children of higher ages. While all selected studies paid attention to elementary-aged students, the inclusion of middle- and high-school students potentially showed a lack of narrowed interest on parental engagement and theoretical perspectives with primary school children.

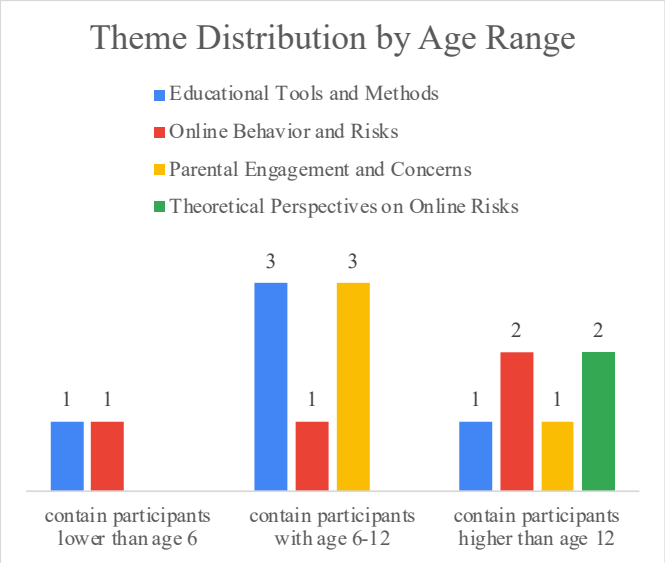


Fig. 6. Theme distribution by age range.

G. Types of Online Risks

While most studies did not gather data on the frequency of specific online risks encountered by children, several highlighted common concerns. Bullying, particularly cyberbullying, emerged as a significant risk mentioned across multiple studies [20], [24], [27], [29], [30], [32]. According to Yusuf et al. [32], 98.3% of surveyed children reported experiencing some form of cyberbullying, underscoring the gravity of this threat for students engaging in online interactions. In contrast, parents displayed a lack of awareness regarding this risk; Larrañaga et al. [27] found that only 6.5% of surveyed parents recognized cyberbullying as a prevalent online risk.

Risks associated with interactions with others online were also noted [18], [20], [21], [24], [28], [29], [30]. Given the differing ages and backgrounds of online contacts, these interactions may involve unwanted behaviors such as sexting and grooming [21], [24]. Furthermore, primary school children may not fully comprehend the implications of their online interactions; for example, researchers found that when discussing privacy, children often did not care who could access or evaluate their information [18]. Many children might have perceived these online connections as mere games, not realizing the real-world consequences of their digital footprints [30].

Concerns about sensitive data also emerged across several studies [20], [21], [26], [28], [31], [32]. These risks primarily revolved around the capacity of applications and websites to safeguard children's data and the children's unawareness of potential data misuse. Although some students recognized threats such as identity theft, primary school-aged children often struggled to respond appropriately to these dangers [28]. For instance, children may inadvertently share personal information without adequately assessing the security measures of websites [31].

Additionally, the concern regarding inappropriate content for children was frequently mentioned [24], [26], [29], [32]. Despite age recommendations for various applications, many children could still access and engage with content beyond their age range [26]. In some cases, primary school students intentionally searched for inappropriate material [30].

H. Methods for Keeping Children Cybersecured

Of the methods highlighted across the 15 studies, two emerged as prevalent: adult supervision and password protection. Adult supervision involved monitoring children's online activities through strategies such as adjusting device settings or placing screens in communal areas [23], [25], [29], [30]. Research indicated that children who are supervised are less likely to experience cyberbullying [23]. However, awareness of this need remained low among parents; only 21.9% of parents surveyed by Abdul Aziz et al. [25] monitored their children's online behavior. Although Annansingh and Veli [29] reported slightly higher rates, only 50.5% of children in their study were supervised. The evidence suggested that insufficient parental supervision persists.

Using passwords to secure devices and information was another commonly mentioned method [22], [26], [30], [31], [32]. In particular, 70.59% of parents surveyed reported using passwords for application installations and device access [26]. Nevertheless, concerns about password practices arose. Conflicting information concerning optimal password strategies was noted [22]; historical recommendations, such as incorporating uppercase and lowercase letters, are no longer endorsed by experts, yet they remain prevalent in educational materials. Thus, both parents and children may lack knowledge

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regarding effective password management. Furthermore, primary school children might not have grasped the importance of keeping passwords confidential [30]. Among the children surveyed by Yusuf et al. [32], 81% had encountered instances of password misuse. Even when children understood that sharing passwords was inadvisable, Witsenboer et al. [31] observed that they often failed to adequately secure their devices.

I. Methods for Teaching Children about Cybersecurity

Among the fifteen studies, three investigated the effectiveness of various learning platforms and activities. The "Would You Rather" activity provided opportunities for teachers to engage students in discussions about hypothetical scenarios concerning privacy and security [18]. This activity reportedly broadened students' perspectives and served as a basis for further educational exploration. One concern raised was the potential for involving sensitive topics, such as the implications of giving house keys to strangers. As noted by one of the participating teachers, clear communication with parents would be essential before implementing such activities in the classroom.

For those seeking independent learning platforms suitable for primary school children, two [19], [20] tested platforms demonstrated effectiveness: "CyberAware" and "Be Smart When Online!". Both platforms aimed to educate students about privacy and security through engaging activities. Specifically, CyberAware was found to be more effective than non-digital educational alternatives [19]. Conversely, "Be Smart When Online!" was shown to be superior to student populations lacking any prior instruction on cybersecurity issues.

IV. Discussion

This scoping review examined the existing literature on cybersecurity education for primary school students, guided by three key research questions. The following discussion synthesizes the findings and explores their implications, addressing each research question in turn.

A. Key Themes in Cybersecurity Education for Primary School Students

As mentioned above, four predominant themes emerged: educational tools and methods, theoretical perspectives on online risks, parental engagement, and student behaviors and online risks. These themes reflect the growing recognition of the multifaceted nature of cybersecurity education for young children, addressing not only technical skills but also the crucial roles of parents and the understanding of online risks, as highlighted in the introduction. The emphasis on tools and methods [18], [19], [20], [21], [22] reflects a pragmatic focus on developing and evaluating interventions. This practical orientation is directly relevant to the pedagogical approaches employed (RQ3), while parental engagement (RQ1) plays a crucial role in mitigating the online risks and threats children face (RQ2). However, the relative scarcity of studies exploring theoretical underpinnings [23], [24] suggests a potential limitation. Grounding interventions in robust theoretical frameworks, such as Protection Motivation Theory [24] or Routine Activity Theory [23], could enhance their effectiveness and generalizability by providing a deeper understanding of the psychological and social factors influencing online behavior. The strong focus on parental engagement [25], [26], [27], [28] acknowledges the crucial role parents play in shaping children's online experiences. However, this also raises questions about the potential burden placed on parents and the need for support and resources to equip them with the necessary knowledge and skills. Finally, the significant attention to online behavior and risks [29], [30], [31], [32] underscores the importance of understanding the specific challenges faced by young children in navigating the digital world.

B. Online Risks and Threats faced by Primary School Students

A significant concern arising from this review is the array of online risks and threats encountered by primary school students, often exacerbated by their developmental stage. Cyberbullying, a pervasive issue across studies, highlights this vulnerability. While reported prevalence rates reach as high as 98.3% [32], the lack of consistent definitions and measurement tools necessitates cautious interpretation. This variation underscores the urgent need for standardized measures specifically designed for primary school children, enabling more accurate prevalence assessments and rigorous evaluation of intervention effectiveness. Furthermore, interventions should move beyond prevention and mitigation of online harassment to cultivate resilience and coping mechanisms in young victims, empowering them to navigate these challenging digital landscapes.

Beyond cyberbullying, interactions with strangers online and exposure to inappropriate content represent significant, interconnected threats. Children's nascent understanding of online privacy and the potential consequences of sharing personal information [18], [31], coupled with a tendency, characteristic of preoperational or concrete operational thinking, to perceive online interactions as mere games [30], amplifies their vulnerability. This digital naiveté, characteristic of their developmental stage, can lead to unintentional disclosures of sensitive information, the formation of risky online relationships, and increased susceptibility to manipulation and exploitation. While this developmental naiveté regarding online interactions can increase vulnerability to manipulation and exploitation, it also presents opportunities for fostering positive online engagement. Effective cybersecurity education must capitalize on children's natural curiosity and enthusiasm for technology while mitigating potential risks. Therefore, effective cybersecurity education must transcend basic technical skills and cultivate a deeper understanding of the social and ethical dimensions of online behavior. This necessitates fostering critical thinking skills, enhancing media literacy, and instilling a sense of responsible digital citizenship. Specifically, educational programs should equip children with strategies for evaluating the credibility of online information, recognizing and responding to potentially manipulative tactics, understanding the permanence of their digital footprint, and navigating online interactions safely and respectfully. Practical pedagogical approaches could include age-appropriate role-playing exercises

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(e.g., simulating scenarios involving online strangers), engaging case studies (e.g., analyzing news stories about cyberbullying), and guided discussions about online ethics and etiquette (e.g., debating the sharing of personal information online).

Applying established theoretical frameworks can provide a deeper understanding of these challenges. Protection Motivation Theory [24], though primarily studied with older populations, offers a lens for understanding how enhancing self-efficacy and emphasizing the perceived severity of online threats can motivate children to adopt safer practices. This aligns directly with the theme of "Student Behaviors and Online Risks," informing the design of interventions to reduce risky online behaviors. Similarly, Routine Activity Theory [23] emphasizes the role of external factors, such as online guardianship and exposure to motivated offenders, in cyberbullying victimization. Connecting this theory to both "Student Behaviors and Online Risks" and "Parental Engagement and Concerns" highlights the importance of parental supervision and limiting opportunities for online victimization. Future research should prioritize the application of these theoretical frameworks specifically to primary school students, investigating how developmental factors influence their applicability and effectiveness in this younger age group. This will provide a stronger theoretical basis for developing age-appropriate and effective cybersecurity education interventions.

C. Effective Pedagogical Approaches and Resources

While analyzing the “Educational Tools and Methods” theme, this review identified diverse pedagogical approaches for enhancing cybersecurity awareness and skills in primary school students. These included digital games (e.g., CyberAware [19]), interactive learning platforms ("Be Smart When Online!" [20]), storybooks [22], and engaging activities like "Would You Rather" [18]. Digital games and interactive platforms hold particular promise due to their potential for immersive learning experiences, increased student engagement, and personalized feedback. Storybooks, while potentially less engaging for older students, can provide relatable contexts and age-appropriate narratives for younger learners. "Would You Rather" activities offer a valuable platform for discussing complex ethical dilemmas and fostering critical thinking skills.

However, these approaches also present limitations. Prior and Renaud [22] highlighted the challenge of maintaining up-to-date information in storybooks and other static resources, particularly concerning rapidly evolving areas like password security. This suggests a need for dynamic, adaptable resources that can be readily updated to reflect current best practices. Furthermore, despite evidence of effectiveness for tools like CyberAware [19] and "Be Smart When Online!" [20], their limited real-world implementation points to a significant gap between research and practice. This gap may stem from factors such as limited teacher training, lack of awareness among educators about available resources, budgetary constraints, or challenges integrating these tools into existing curricula. Moreover, the review revealed a scarcity of research investigating the long-term impact of these pedagogical approaches on students' cybersecurity knowledge, skills, and behaviors.

D. Children's Attitude Towards Cybersecurity

Reflecting upon the findings of the articles, this review revealed a nuanced understanding of children's attitudes towards online safety. While several studies indicated a growing awareness of online risks among primary school students [29], [32], this awareness appears superficial and fragmented rather than comprehensive. For example, students could identify inappropriate content [32] and demonstrate awareness of unsuitable online games [29]. However, this nascent understanding can paradoxically foster overconfidence in their abilities. Witsenboer et al. [31] found that many children displayed an optimistic, even unrealistic, perception of their digital literacy, leading them to underestimate the risks associated with sharing personal information. This disconnect between perceived and actual competence highlights a critical challenge in cybersecurity education. As Panskyi and Korzeniewska [21] suggest, younger children's limited capacity for abstract thinking and risk assessment, a characteristic of their developmental stage, contributes to a sense of invulnerability, increasing their susceptibility to online manipulation.

These findings suggest that simply informing children about online risks is insufficient. Effective cybersecurity education must move beyond basic awareness and cultivate critical thinking skills. By actively engaging children in analyzing online behaviors and evaluating potential consequences, we can empower them to make informed decisions and develop adaptive strategies for navigating the complex digital landscape. This necessitates a shift towards pedagogical approaches that prioritize critical thinking, problem-solving, and responsible digital citizenship [cite examples of such approaches if discussed in the reviewed literature or other relevant sources]. This underscores the need for future research to investigate the efficacy of interventions specifically designed to foster these higher-order thinking skills in the context of online safety. Furthermore, research exploring the developmental trajectory of online risk perception and its relationship to cybersecurity behaviors in primary school children would be valuable.

E. Stakeholder Collaboration and Communication

While analyzing the results, a significant disconnect among key stakeholders--parents, teachers, and policymakers--regarding children’s digital safety became evident.. This misalignment hinders the effective implementation of cybersecurity education. While innovative educational tools and platforms are emerging [19], [20], a critical discrepancy exists in perceptions of responsibility. For instance, Abdul Aziz et al. [25] found that 80% of surveyed parents believed schools should provide cyber ethics lessons, contrasting sharply with the views expressed by teachers in Martin et al. [30], who often considered this a primarily parental duty. This lack of consensus underscores a systemic issue in how responsibilities are assigned and understood among stakeholders, potentially leading to gaps or duplication of efforts.

Furthermore, the communication breakdown extends to the dissemination of effective cybersecurity practices. Prior and

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Renaud [22] highlighted parental misconceptions regarding current best practices for online safety, particularly in areas like gaming and online interactions. This knowledge gap can create a false sense of security among parents [27], who may overestimate their ability to address evolving online threats. Such inflated confidence can be detrimental, leaving children vulnerable to risks that surpass basic parental controls. Effective cybersecurity education requires a concerted effort, and this disconnect among stakeholders undermines the development of a cohesive and comprehensive approach. Establishing clear communication channels and shared understanding of best practices is crucial to bridging this gap and fostering a truly collaborative approach to children's online safety. This could involve developing joint training programs for parents and teachers, creating accessible online resources, or establishing school-community partnerships focused on digital literacy and safety. Further research is needed to explore effective strategies for fostering communication and collaboration among stakeholders and evaluating the impact of such interventions on children's online safety outcomes.

V. Limitations, Conclusions, Implications

This review has several limitations. First, the search was limited to Scopus and Google Scholar, potentially overlooking relevant articles indexed in other databases. Second, the article selection process relied on title keywords to identify studies focusing on primary school students, which may have excluded relevant articles that did not use these specific terms. Third, the 10-year timeframe, while capturing recent research, might not fully reflect the evolving landscape of cybersecurity education, particularly given the significant shifts in technology and policy following the COVID-19 pandemic. These limitations could have introduced bias and impacted the comprehensiveness of the review. Future research could expand the database search, refine the inclusion criteria, and analyze trends over different periods to address these limitations and gain a more complete understanding of cybersecurity education for primary school students.

In conclusion, this scoping review of 15 studies highlights the critical need for enhanced cybersecurity education tailored to primary school students. The analysis revealed four key themes: the development and application of educational tools, theoretical frameworks for understanding online risks, parental engagement, and children's online behaviors and the risks they encounter. While promising pedagogical approaches exist, including digital games and interactive platforms, their effectiveness is often hampered by limited implementation and outdated resources. Furthermore, a significant disconnect among parents, teachers, and policymakers regarding responsibility and best practices hinders the development of a comprehensive approach to cybersecurity education.

The findings have some implications for future research in cybersecurity education for primary school students: (1) Teacher-Parent Communication: Investigate effective communication strategies between teachers and parents regarding children's online activities and safety. This includes exploring methods for sharing information about online risks, promoting open dialogue, and addressing potential discrepancies in perceptions of responsibility. (2) Collaborative Interventions: Develop and evaluate collaborative interventions involving teachers, parents, and children. These interventions should focus on building shared understanding, promoting joint problem-solving, and creating safer online environments through unified strategies. (3) Policy and Resource Provision: Explore the role of policy in supporting and facilitating communication and collaboration among stakeholders. This includes examining how policy can enhance resource provision, promote best practices, and ensure equitable access to cybersecurity education for all students. (4) Cyberbullying Awareness and Prevention: Develop and test interventions specifically targeted at increasing parental awareness of cyberbullying risks and promoting effective prevention strategies. This includes exploring the effectiveness of educational campaigns, parent training programs, and school-based initiatives. (5) Holistic Understanding of Children's Online Experiences: Conduct research that takes a holistic perspective, examining the interplay between children's online experiences at home, in school, and in other contexts. This includes investigating the influence of peers, media, and other social factors on children's online behavior and risk perception.

Last but not least, the findings of this review have some implications for practical improvements. First, schools should prioritize establishing robust communication channels with parents. Regular workshops or information sessions can foster a shared understanding of cybersecurity challenges, practical strategies, and respective roles in promoting online safety. These sessions could cover topics such as age-appropriate online activities, parental control tools, recognizing and responding to cyberbullying, and promoting responsible digital citizenship. Second, educators should actively evaluate and integrate available cybersecurity resources into the curriculum. Numerous engaging and effective tools exist but remain underutilized. Incorporating these resources can create more interactive and relevant learning experiences, equipping students with essential cybersecurity skills. This might involve integrating cybersecurity modules into existing subjects, leveraging game-based learning platforms, or incorporating real-world case studies. Finally, fostering a collaborative learning environment where students can actively discuss and practice cybersecurity strategies is essential. This holistic approach, combining parent-school communication, effective resource integration, and collaborative learning, can empower students to navigate the digital world safely and confidently, promoting their overall online well-being. By prioritizing these practical steps, stakeholders can create a supportive ecosystem that nurtures responsible digital citizens equipped to thrive in today’s interconnected world.

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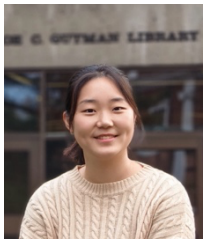
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Yukun Xu, born in Shenzhen, China, earned a Bachelor of Science in Elementary Education and Child Development from Vanderbilt University in 2024. Currently pursuing a Master's in Learning Design and Technology at Stanford University, she brings experience from roles as an Assistant Teacher at the Vanderbilt Child and Family Center (May-August 2022), caring for children aged 6 months to 5 years and designing developmentally appropriate activities, and as an IT Intern at Lavner Education (June-August 2023), where she taught children aged 5-12 to use technology for creative projects like video editing and explored technology's application in fields like medicine and fashion.



Hui Philip Li, PhD, born in China, earned his PhD in Education from the University of Hong Kong (HKU) in 2000. He holds the Chair Professorship in Early Childhood Education at The Education University of Hong Kong and has served as Dean of the Shanghai Institute of Early Childhood Education (Shanghai Normal University), Program Director (Master's) at Macquarie

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University, and Program Director (MEd) at HKU. He also holds honorary professorships at Macquarie University and Shanghai Normal University, and is a Visiting Professor at several mainland Chinese universities. As the founding editor of *AI, Brain & Child* (a Springer-Nature journa), Prof. Li's extensive editorial experience includes Co-Editorship of the *Journal of Research in Childhood Education* and Associate Editorship of four SSCI journals (*Early Education and Development*, *Early Child Development and Care*, *Frontiers in Psychology*, and *Sustainability* [education section]), as well as editorial board membership in six international peer-reviewed journals. His research has been funded by sources including the National Natural Science Foundation of China (NSFC), the General Research Fund (Hong Kong), the Australian Children's Education and Care Quality Authority, and the Quality Education Fund. Since 2021, he has been consistently ranked among Stanford University's World's Top 2% Scientists.

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APPENDIX A

TABLE III
INFORMATION OF 15 ARTICLES REVIEWED IN THIS STUDY

Author/year (Country)	Participants' Details				Research Design		
	Sample Size (Children)	Sample Size (Parents)	Sample Size (Other Stakeholders)	Relevant Children's Age	Collection Method of Data	Mentioned online risks	Purpose of Study
Abdul Aziz et al., 2023 (Malaysia)		32		12 and below	Qualitative	Misuse of personal data	Measure of child's digital use (from parents' perspective)
Aizenkot, 2022 (Israel)	Elementary: 4077 Middle School: 1279 High School: 363			Elementary: 3rd - 6th grade Middle School: 7th - 9th grade High School: 10th - 12th grade	Quantitative	Bullying	Test of existing theory
Alashwali & Alashwali, 2022 (Saudi Arabia)		119		6-12	Quantitative	Inappropriate content (children using apps that are not recommended for their age); Misuse of personal data	Measure of child's digital use (from parents' perspective)
Annansingh & Veli, 2016 (UK)	300	300	13	7-11	Mixed	Interaction with people (interaction with other gamers and people on social networks can expose children to inappropriate content for their age); Inappropriate content (age-restricted games, inappropriate conversation with others); Bullying	Measure of child's digital use (from children's and parents' perspective); Risks as identified by teachers
Blinder et al., 2024 (USA)	21		13	7-12	Qualitative	Misuse of personal data (privacy and security in both online and off-line environments)	Measure of children's perception of privacy issues
Giannakas et al., 2019 (Greece)	52			9-12	Quantitative	Misuse of personal data (privacy and security)	Test of a proposed learning platform

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CYBERSECURITY MATTERS

Author/year (Country)	Participants' Details				Research Design		
	Sample Size (Children)	Sample Size (Parents)	Sample Size (Other Stakeholders)	Relevant Children's Age	Collection Method of Data	Mentioned online risks	Purpose of Study
Larrañaga et al., 2022 (Spain)		850		9-12: 434 families 13-17: 416 families	Quantitative	Bullying	Measure of parents' awareness and attitude toward bullying
Martin et al., 2023 (USA)			10	5-11	Qualitative	Inappropriate content (such as nudity); Interaction with people (children treating interaction with strangers as games or interactions with peer, not taking it seriously); Bullying; Misuse of personal data (sharing passwords with friends and personal information on websites; not understanding digital footprint)	Measure of teachers' perception towards risks that children face
Nicolaidou & Venizelou, 2020 (Cyprus)	73			6th grade	Mixed	Misuse of personal data; Interaction with people (hackers); Bullying	Test of a proposed learning platform
Panskyi & Korzeniewska, 2023 (Poland)	243			8-10	Quantitative	Misuse of personal data (viruses); Interaction with people (threats and dangers, groom, etc.); Inappropriate content (dangerous content online)	Measure of children's awareness of online risks and their perceptions of their awareness
Prior & Renaud, 2020 (UK)		12	8	4-9	Qualitative	Misuse of personal data (focusing mostly on passwords); Interaction with people (stranger danger)	Measure of commonly-taught password practices; Proposal of age-appropriate, up-to-date password ontologies
Quayyum, 2025 (Norway)	10			9-12	Qualitative	Misuse of personal data (privacy and security in both online and off-line environments)	Measure of children's perception of privacy issues

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Author/year (Country)	Participants' Details				Research Design		
	Sample Size (Children)	Sample Size (Parents)	Sample Size (Other Stakeholders)	Relevant Children's Age	Collection Method of Data	Mentioned online risks	Purpose of Study
Teimouri et al., 2018 (Malaysia)	420			9-16	Quantitative	Interaction with people (intimate relationship); Misuse of personal data; Inappropriate content (sexual or violent content); Bullying	Test of existing theory
Witsenboer et al., 2022 (Netherlands)	Elementary: 140 High School: 96			Elementary: 9-13 High School: 15-19	Mixed	Misuse of personal data (sharing personal information, passwords); Interaction with people (social media, dangerous websites)	Measure of children's awareness of online risks and their behaviors
Yusuf et al., 2023 (Malaysia)	420 from Primary and Secondary School			Mean = 12, 74% are between 9 to 16	Quantitative	Misuse of personal data; Interaction with people (sexting and conversations about sex); Inappropriate content (such as nudity); Bullying	Measure of children's online experience

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APPENDIX B

TABLE IV

FINDINGS OF 15 ARTICLES REVIEWED IN THIS STUDY

Author/year (Country)	Research Result			Limitations
	Methods to Keep Children Cybersecured	Methods to Teach Cybersecurity	Interesting Findings	
Abdul Aziz et al., 2023 (Malaysia)	Parental guidance is a current method implemented by parents, by placing devices such as television or computer in public space at home (eg. Living room).		Parents (80%) suggests for schools to provide cyber ethics education to parents and children. Parents also asked for governments to censor websites.	This study only interviewed parents about their thoughts on their children's digital use and what's needed. There's no measurement from children or other perspectives.
Aizenkot, 2022 (Israel)	Online guardianship (presence of people that may intervene in face of bullying) decreases the possibility of being bullied. If children have more reliable relationship off-line (in which they trust), the possibility of being bullied also decreases.			The theory tested (Routine Activity Theory) only explained for 8.3% of the variance in children's likelihood of being bullied, which means that there are other factors that are influential but not tested in this study. The theory was also originally designed for off-line environment.
Alashwali & Alashwali, 2022 (Saudi Arabia)	Parental guidance is demonstrated through parents deciding whether or not to download certain apps (57.14%). Few parents (11.76%) would research the app before downloading. Parents also talk about privacy concerns with children, with only 7.56% of parents who don't talk about it at all. Passwords are used by 70.59% of the parents on their children's device, whether to control logging into the device or app installation.			This study tested mainly parents that are highly educated, female, and from the Western region of Saudi Arabia, which limits the type of family this study may collect data about.
Annansingh & Veli, 2016 (UK)	Parental guidance is lacking, as teachers point out that parent's unawareness of children being online. Parents are also unaware of children's gaming experience online. Children are also not supervised while being online (only 50.5% of the children are supervised). Filters are used, but these "can be easily altered by a child."	The school teaches students about e-safety.	Children understands the physical risks (such as inappropriate content) of the internet more than the emotional ones (such as bullying). Teachers in this study also mentioned the importance of parents being educated on e-safety measures.	This study provided few data collected from parents and children and on the type of questions they asked, which can effect how one may interpret the responses.

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Author/year (Country)	Research Result			Limitations
	Methods to Keep Children Cybersecured	Methods to Teach Cybersecurity	Interesting Findings	
Blinder et al., 2024 (USA)		Schools can use Would You Rather activities to open up students' perspectives and talk about privacy and security in class. Children also learn from parents' mistakes: children in this study were observed talking about how dad lost car key once and what strategies dad took against it.	Would You Rather activities can serve as extension activities for student learning. Teachers in this study raise the concern of communicating with home before the activity takes place due to the potential controversial nature of the questions. Children generally don't "care" about who might have access to or judge their information. Children generally follow the rules set up by adults nd enjoy the security it comes with; they are not usually rule breakers.	This study did not tested for change in students' perceptions of things before and after the activity.
Giannakas et al., 2019 (Greece)		CyberAware is an app that offers self-paced learning digitally. Learning was assessed by comparing the performance of students who used this app and that of students who worked on a non-digital activity that is similar to the app. Both groups of students received traditional course on privacy and security before engaging in any forms of activities and before taking any tests.		The information taught by this platform contradicts the good password practices proposed in Prior & Renaud (2020): this may offer some questions about the "correctness" of the information and strategies taught through this platform. Because of the set up of this study, it is unclear whether the increase in performance is due to the presence of feedback in the tested activities. Specifically, the non-digital learning activity seems to lack any form of feedback, whereas CyberAware offers feedback on learners' answers.
Larrañaga et al., 2022 (Spain)	Parental awareness is lacking: parents are generally unaware of bullying as a risk that their children face. In families with children between the age of 9 and 12, only 6.5% of the parents perceived bullying as risk. Moreover, only 3.3% of the parents perceived bullying as a perpetrated risk.		Parents seem to be overconfident in bullying issues. In families with children between the age of 9 and 12, 81.8% of the parents believed in their capability in helping their child. Additionally 22.5% of the parents think their child is resilient and can deal with bullying.	This study relied mostly in parent reports in bullying issues without presenting data obtained from the children's perspectives.

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Martin et al., 2023 (USA)	Teachers in this study mentioned the lack of parental supervision of children's digital use at home. Digital experiences at schools are safeguarded by firewalls and filters. Teachers also monitor children's online activities.	Lessons and discussions were provided at school to educate children about safe online practices. Children were taught to report to guidance counselors when facing bullying issues.	Children at this age don't quite know what can be harmful to them. Teachers in this study also suggested instructions for parents on digital safety.	This study solely relied on interviews of teachers, without the perspectives of children or parents in the stated environment. The generalizability of this study is also limited because data was collected within a single state in the U.S.
Nicolaidou & Venizelou, 2020 (Cyprus)		"Be smart when online!" offers a learning platform for children to independently learn about e-safety issues informally. The platform was tested with two groups of learners, those who used this platform and those who didn't. Equivalence in performance was established between the two groups before the learning platform was introduced.		Although the study incorporates triangulation of data, collecting both test data and interview data, the interview was only conducted with children who had used the learning platform.
Panskyi & Korzeniewska, 2023 (Poland)		COVID, which may be associated with children's increase use of internet, increased children's awareness of their digital security skills and performance in recognizing and solving digital security problems.	Children may not always be aware of what they know or don't know.	This study didn't measure for specific factors of COVID that may have contributed to the change in children's performance.
Prior & Renaud, 2020 (UK)		Storybooks are used to teach children about good password practices.	Books and websites contain information that were ambiguous, out-of-date, or incorrect about good password practices. Children are also taught about "stranger danger". However, danger can also come from friends and people that children know.	This study didn't focus on measurement and analysis on online risks. It focused mostly on passwords and good practices surrounding passwords.
Quayyum, 2025 (Norway)	Children perceive adults, such as parents and police, as people for help and advice. Children understand passwords to be a method to protect their information.		Children understand risks such as identity theft. Children perceive strangers as a source of danger. Children also understand that "defying parents can lead to negative consequences."	This result of this study can hardily be generalized. The focus of the study is mainly in proposing a method through which adults can understand how to help children.

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Teimouri et al., 2018 (Malaysia)	Increase of children's perceived severity of online risks decreases possibility of children's exposure to online risks. Increase of children's digital literacy (children's ability in protecting self and perform tasks online) increases children's exposure to online risks.			This study used convenient sampling, which limits the generlizability of the study. This study also measured based on self-reported survey from children, which constrains "online risks" as what is interpreted by the children. The survey used in this study was also adapted from U.S. and Europe, and certain wordings were changed, which may impact the effectiveness of measurement.
Witsenboer et al., 2022 (Netherlands)	Children understand that passwords should not be shared, but they also don't lock their device sufficiently.	Children mentioned learning about and developing online secure behaviors at school, whether through learning about it or seeing others experiencing the negative consequences. Children also express interest in learning about cybersecurity in the last years of elementary school.	Children may be unaware of the danger behind websites. Results show that they simply put in their personal information without double-checking the websites. Children are overconfident about their cybersecurity knowledge.	This study didn't interview the elementary students. Instead, it provided interview data of high school students, asking them to reflect on their elementary years. This may offer misleading information on primary school experience. This study was also conducted at one school, limiting its generalizability.
Yusuf et al., 2023 (Malaysia)			Online risks are prevalent: 91% of children were asks by someone to do inappropriate acts, and 98.3% of children experienced modest level of bulllying.	This study didn't target and assess specific age groups. This study also solely measured the risks children faced, without much focus on methods that children and others take to counter these risks.

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